Part 4: Practice Assignments

1. $$\lim_{x \to 4} (x + 6)$$
2. $$\lim_{x \to 3} (4x)$$
3. $$\lim_{x \to 2} (x^2 + 1)$$
4. $$\lim_{x \to 10} \left(\frac{x + 2}{3}\right)$$
5. $$\lim_{x \to 9} \sqrt{x}$$
6. $$\lim_{x \to 0} (3x^2 + 8x - 2)$$
7. $$\lim_{x \to 100} 15$$
8. $$\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$$ *(Hint: Use the Difference of Squares rule!)*
9. $$\lim_{x \to 1} \frac{x^2 - 1}{x - 1}$$ *(Hint: 1 is a perfect square because $1 \times 1 = 1$)*
10. $$\lim_{x \to 3} \frac{4x - 12}{x - 3}$$ *(Hint: Pull out the Greatest Common Factor from the top!)*

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11. The Rocket Launch A model rocket is launched, and its altitude in feet is measured by the equation $Altitude = x^2$ (where x is seconds). Set up a limit and use the factoring trick to find the rocket's exact speed at exactly **4 seconds**. *(Hint: Set up the limit as $x \to 4$, use x^2 for the moving distance and 4^2 for the fixed distance).*
